Case Study II: Styrian Health Data

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Random Forest and Selected Hypotheses

- Applied **Random Forest** to analyze blood pressure
  - H1: Can we statistically prove the relationship between age and BP?
  - H2: Are there differences in blood pressure by gender, diabetes, smoking, cholesterol, well-being?

Final Dataset

- **Original rows:** 16,386
- **Rows after removing nulls:** 16,004 (removed 382)
- **Rows after filtering age 13–100:** 15,219 (removed 785)
- **Total removed:** 1,167

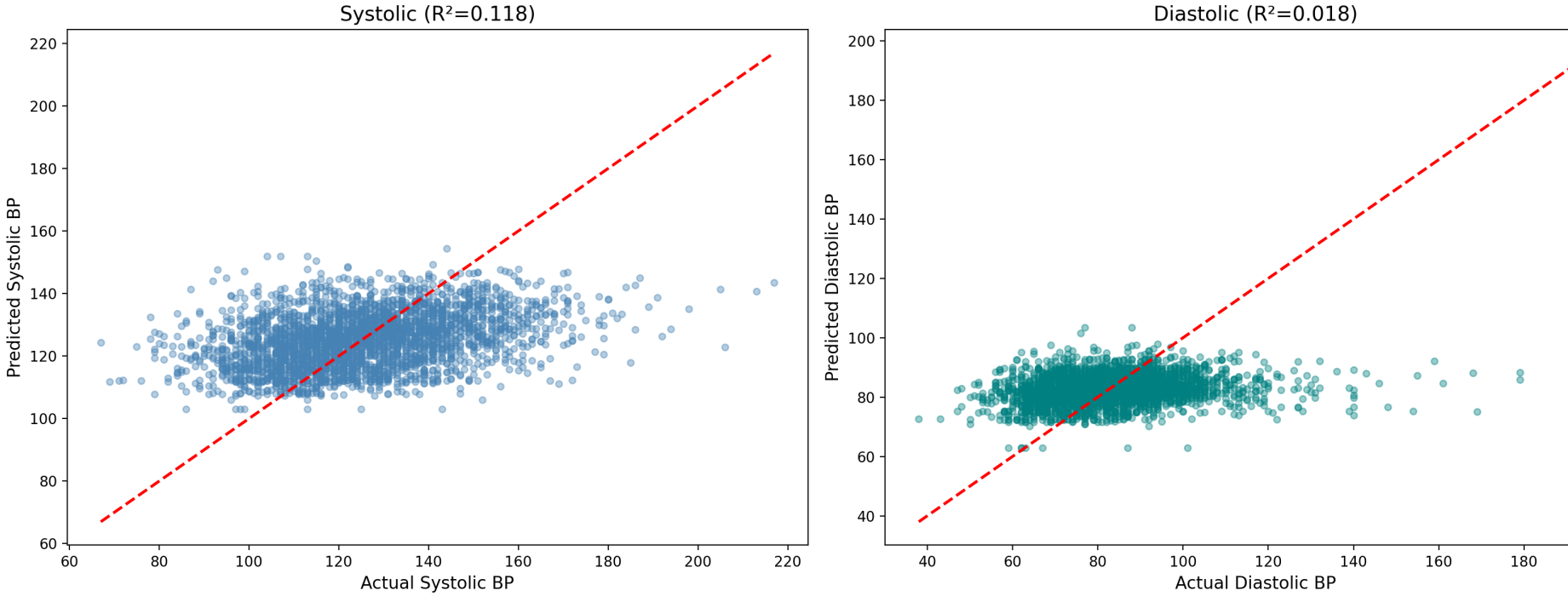
Data Preparation for Random Forest

- Random Forest requires all input features to be numeric
  - Converted categorical/boolean columns to integers

Random Forest — Blood Pressure Prediction

- Multi-output Random Forest model to predict **Systolic and Diastolic BP**
- **Features used:** Age, Gender, Smoker, Known Blood Sugar, Known Cholesterol, Under Treatment, Health Condition
- **Hyperparameters:**
  - n_estimators=500, max_features='sqrt'
  - min_samples_leaf=5, max_depth=None
  - random_state=42, n_jobs=-1

Random Forest — Model Evaluation

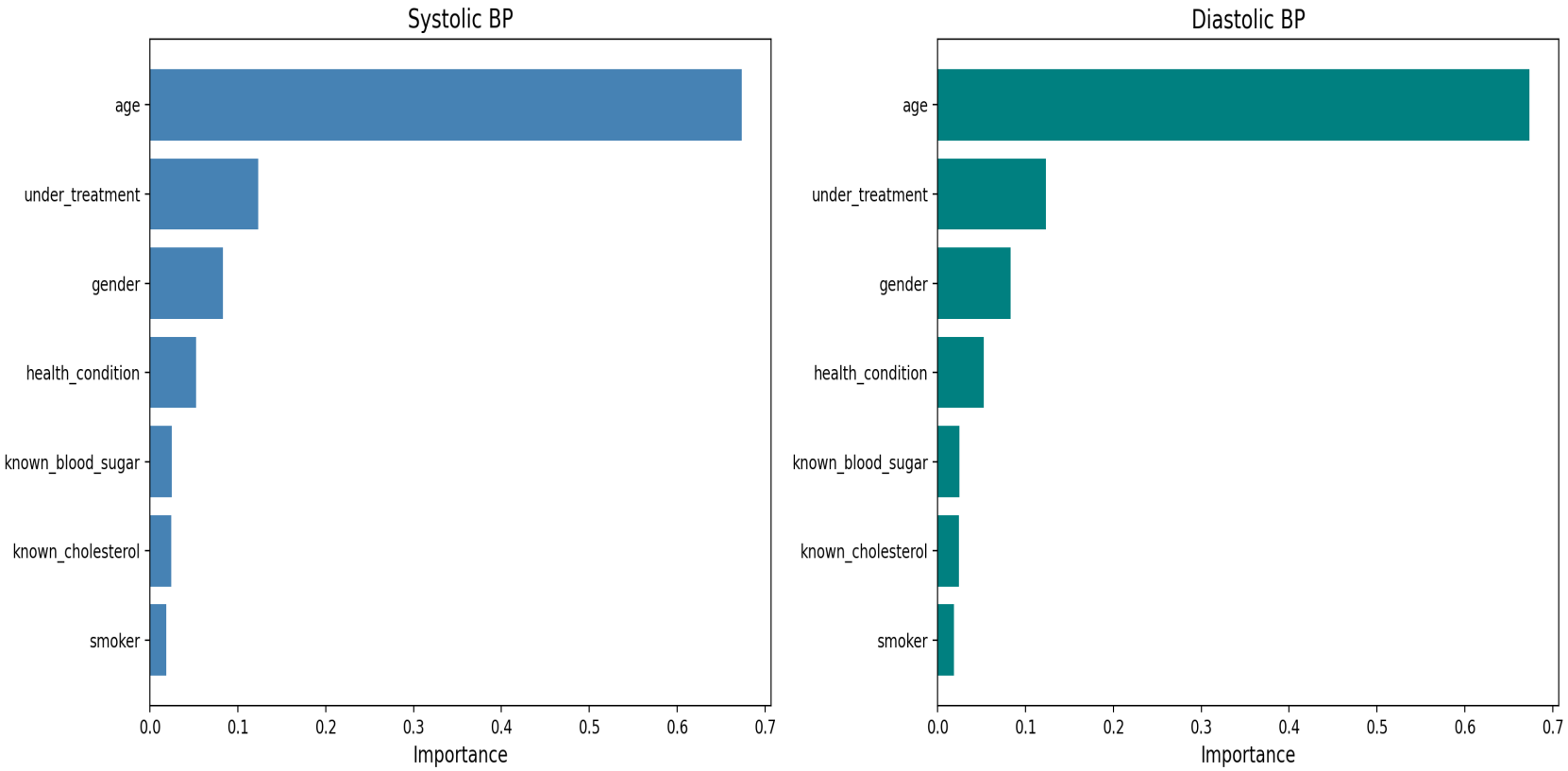


- Systolic BP predictions follow a weak positive trend
- Diastolic BP predictions are very noisy

Random Forest — Model Evaluation

- **Systolic BP:**
  - $R^2 = 0.118$ — low, model explains only 12% of variance
  - MAE = 14.05 mmHg, RMSE = 17.98 mmHg
- **Diastolic BP:**
  - $R^2 = 0.018$ — almost no predictive power
  - MAE = 10.70 mmHg, RMSE = 14.43 mmHg
- Observations:
  - Overall model fit is weak — variance in BP is high
  - RMSE $>$ MAE indicates presence of outlier predictions

Feature Importance — BP Levels



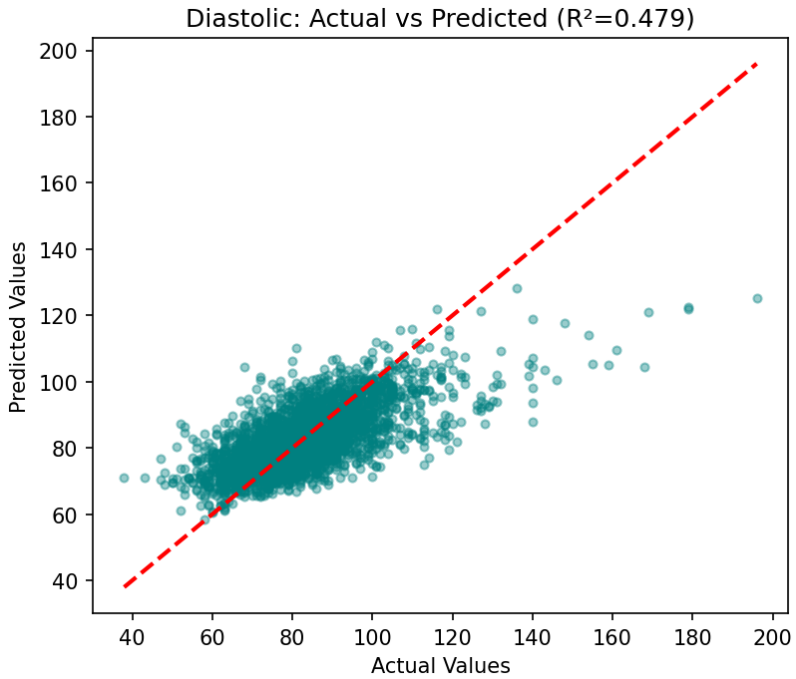
Feature Importance — BP Levels

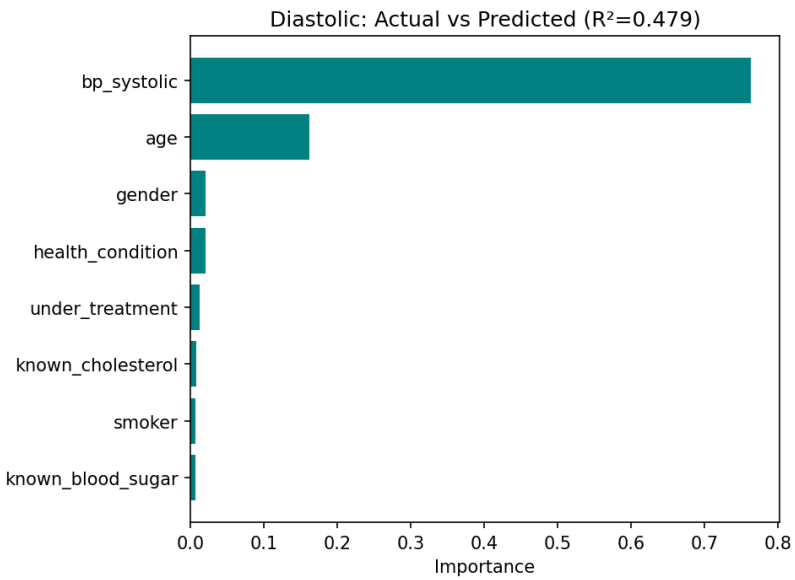
- **Age** is the dominant feature in both models (importance $\approx$ 0.67)
- Followed by **Under Treatment**, **Gender**, and **Health Condition**
- Smoker, Blood Sugar, and Cholesterol contribute least

Overall Model Performance

- Model has **weak predictive power** for both BP types
- **Age** alone explains most of what the model can predict
- High MAE suggests predictions can be off by **10–14 mmHg** on average
- This is clinically significant — a 14 mmHg error matters in practice

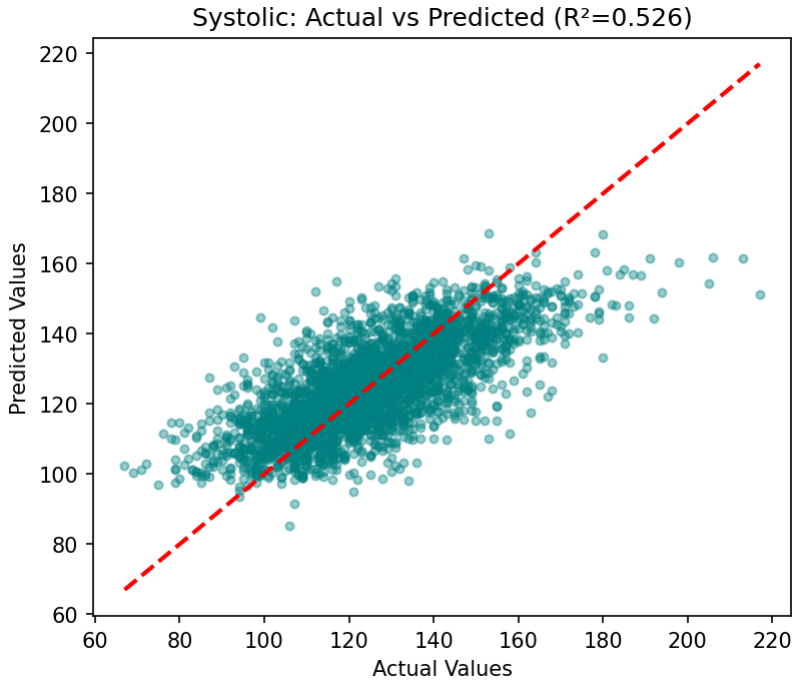
Diastolic BP — Single Output Model

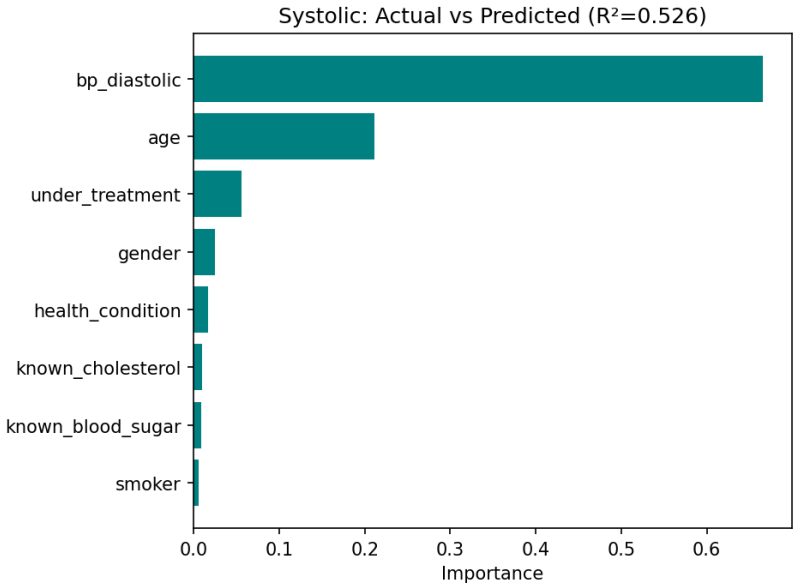




- $R^2 = 0.479$, MAE = 7.73 mmHg, RMSE = 10.52 mmHg
- **Systolic BP** dominates as the strongest predictor (0.76)
- Age follows as second most important feature (0.16)

Systolic BP — Single Output Model

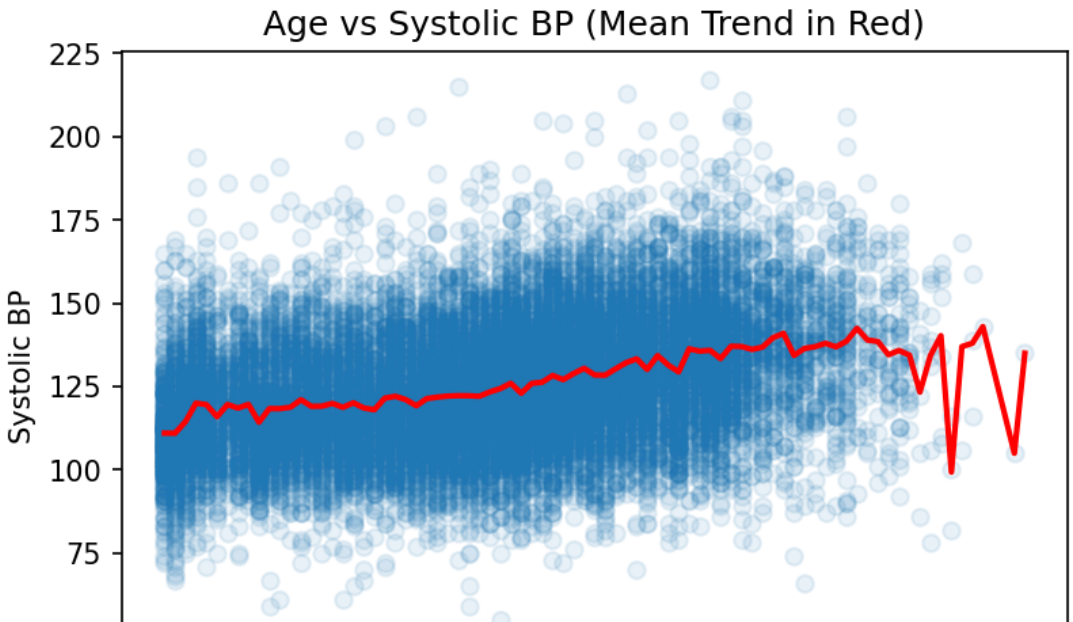




- $R^2 = 0.526$, MAE = 10.31 mmHg, RMSE = 13.18 mmHg
- **Diastolic BP** dominates as the strongest predictor (0.67)
- Age follows as second most important feature (0.21)

Hypothesis 1 — Age vs Blood Pressure

- H1: Can we statistically prove the relationship between age and BP?
  - Age is the most important predictor for both Systolic and Diastolic BP
  - Feature importance: Age = 0.673



Hypothesis 2 — Other Factors vs Blood Pressure

- H2: Are there differences in BP by gender, diabetes, smoking, cholesterol, well-being?
  - Under Treatment: 0.123 — highest among lifestyle factors
  - Gender: 0.083
  - Health Condition: 0.053
  - Known Blood Sugar: 0.025
  - Known Cholesterol: 0.024
  - Smoker: 0.018 — least influential
- All factors show **low importance** compared to Age
- R^2 very low — model cannot distinguish meaningful BP differences by these factors alone